

Student Performance Predictor

Project Report | Fundamentals in AI ML

Student Information

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1 Objective

To design a simple machine learning model that predicts whether a student will **Pass** or **Fail** based on their marks, attendance, and average daily study hours using basic Python and `scikit-learn`.

2 Introduction

This project applies **Logistic Regression** to predict student performance. By leveraging a synthetic dataset generated programmatically, the model is fully self-contained. Logistic regression is an ideal entry point for classification tasks within the AI/ML domain, providing insights into feature influence on binary outcomes.

3 Methodology & Algorithm

The implementation follows a systematic pipeline:

- **Data Generation:** Creating a random dataset representing marks, attendance, and study hours.
- **Labeling:** Classifying students as 1 (Pass) or 0 (Fail) based on a weighted scoring mechanism.
- **Data Splitting:** Dividing the data into training and testing subsets (e.g., 70/30 split).
- **Training:** Fitting a Logistic Regression model to the training data.
- **Evaluation:** Assessing the model using accuracy scores and classification metrics.

4 Tools & Technologies

- **Programming:** Python 3.x
- **Libraries:** `pandas` (Data handling), `scikit-learn` (ML training & evaluation), `random` (Synthetic data generation).
- **IDE:** Visual Studio Code

5 Model Evaluation

The model was tested against a validation set of 300 samples.

Key Metric

Overall Accuracy: 0.8667 (86.67%)

5.1 Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Fail)	0.4000	0.0513	0.0909	39
1 (Pass)	0.8746	0.9885	0.9281	261
Weighted Avg	0.8129	0.8667	0.8192	300

6 Example Data Snapshot

Below are the first 5 rows of the generated dataset used for training:

Marks	Attendance	Study Hours	Result (Pass)
48.72	95.58	4.10	1
79.96	96.42	6.22	1
69.24	74.73	7.35	1
42.41	77.55	6.74	1
56.32	63.36	8.81	1

7 Result & Conclusion

The model successfully achieved an accuracy of **86.67%**. While it is highly effective at identifying passing students, the recall for the 'Fail' class suggests room for improvement via feature engineering or handling class imbalance.

This project demonstrates core machine learning principles—data preprocessing, model selection, and evaluation—using standard Python tools. It serves as a foundational prototype for more complex predictive analytics in educational environments.